**DO NOW — RECALL**

# Lesson 055 · Choosing Data Structures for Problems

Name: _____ Class: _____ Date: _____

**1. What is a hash collision?**

**2. When can linked-list insertion be O(1)?**

**3. What is a BST's advantage when balanced?**

**4. An empty queue receives 4, 8 and 2 in that order. Repeatedly dequeue an item and push it onto an empty stack until the queue is empty. Trace the sequence returned by popping the stack until it is empty.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Two different keys map to the same bucket.

### 2. Model answer

When a reference to the insertion node is already known.

### 3. Model answer

Ordered search can require O(log n) comparisons.

### 4. Model answer

2, 8, 4. The queue supplies 4, 8, 2; the stack reverses that order.

### By the end, you can…

• Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

• Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

• Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

### Vocabulary — know these terms

operation — Action such as lookup or insertion.

random access — Direct access by numeric index.

trade-off — A benefit weighed against a cost.

complexity — Growth in resource use.

NEW LEARNING

# Read, annotate and explain

### Discover

Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

### Trace

Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

### Evaluate

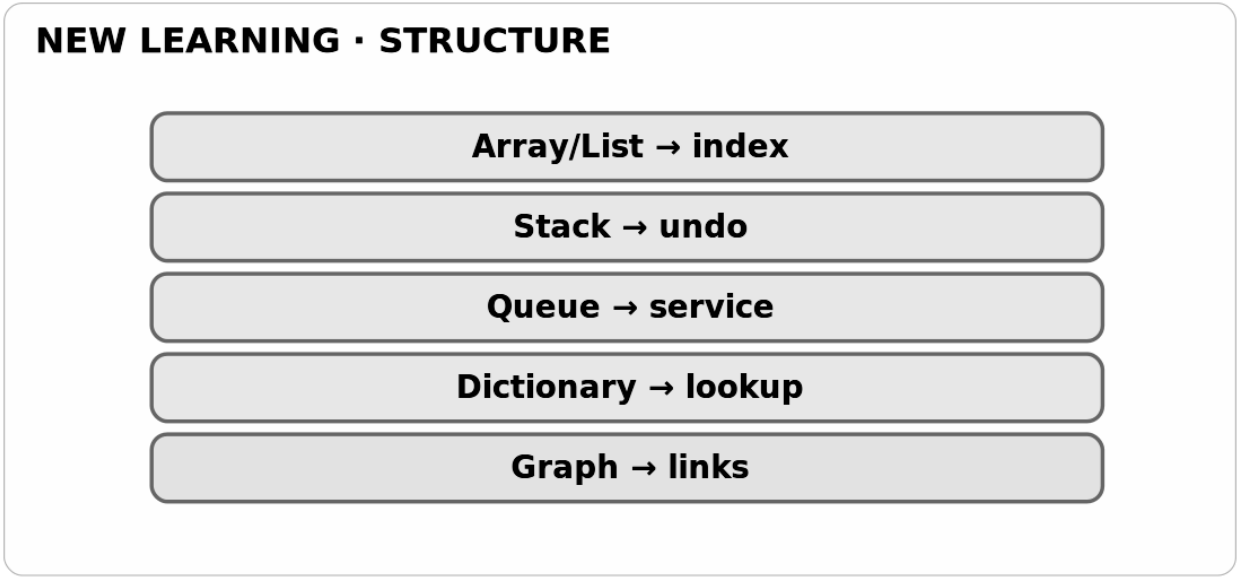
Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




NEW LEARNING · STRUCTURE: Array/List → index, Stack → undo, Queue → service, Dictionary → lookup, Graph → links

I DO — WORKED EXAMPLE

# Follow the worked method

**Choose a structure for a browser's Back action and justify.**

**Explain the trace**

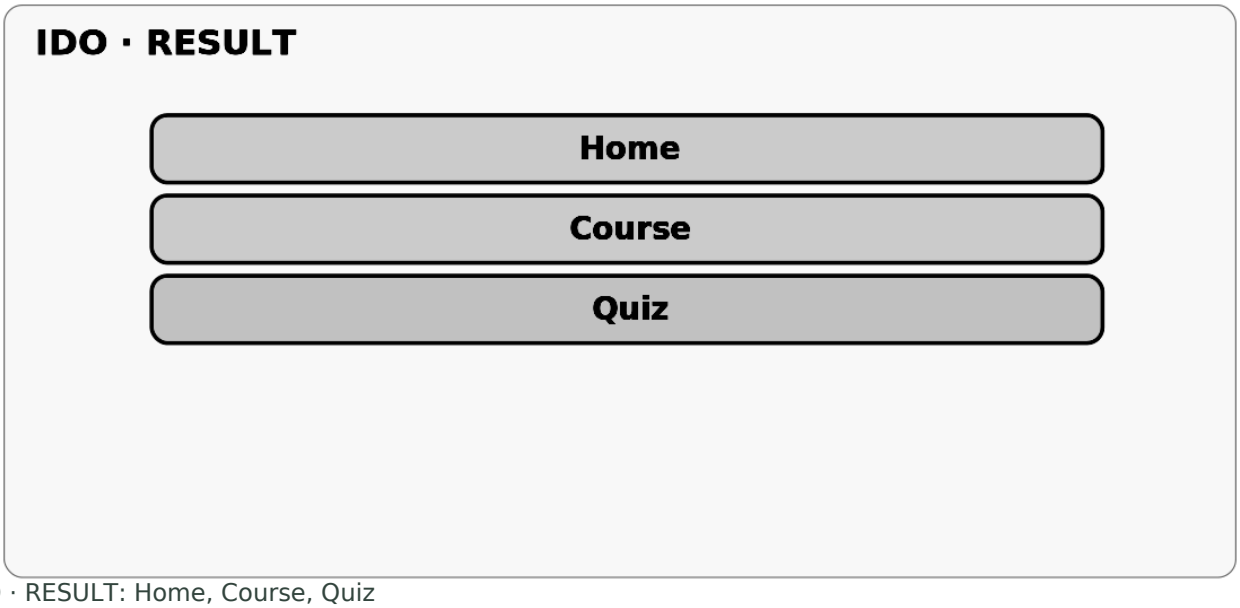
Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Choose a structure for a browser's Back action and justify.

Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.




IDO · RESULT: Home, Course, Quiz

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Choose a structure for clinic tickets processed in arrival order.
Source code / input:
queue.Enqueue(ticket)
next=queue.Dequeue()

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Match an undo feature, printer queue, pupil ID lookup and road map to a structure. [4]

### Question 2

A timetable stores fixed grid of rooms and periods. Recommend a 2D array or graph and explain operations. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Evaluate using List<T> versus linked list for a million-item collection, focusing on frequent random index access versus middle insertion with a known node. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**The same data structure is always optimal for every collection.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

**WE DO — CHECK AND IMPROVE**

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

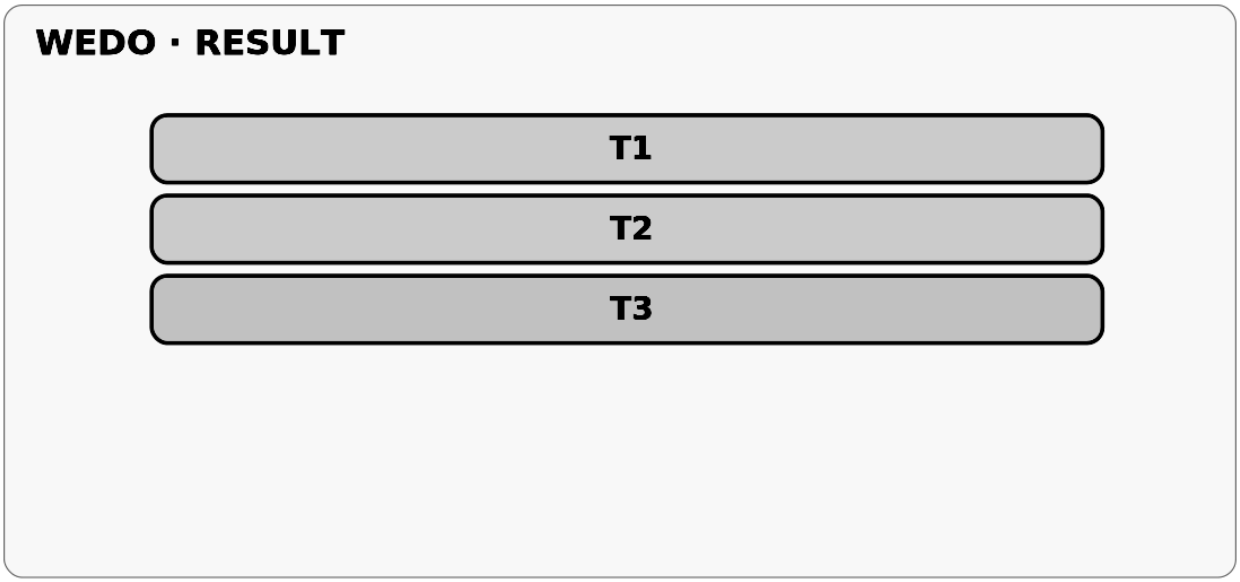
Explain the trace
Queue FIFO serves earliest arrivals first, subject to real-world priority policies.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Choose a structure for clinic tickets processed in arrival order.

Queue FIFO serves earliest arrivals first, subject to real-world priority policies.




WEDO · RESULT: T1, T2, T3